

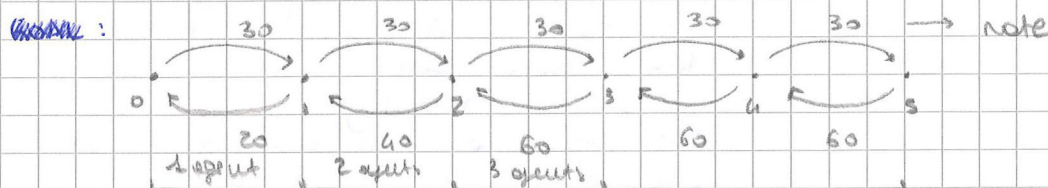
QUEUEING PROCESSES

EX: Call center

- 30 call arriving per hour
- 3 agents reply and take 3 min per call (20 call per hour)
- if the agents are busy, customers wait (up to 2)
- the rest are lost, call another time

=> INTERVAL TIMES: incoming call every now and then
time between calls

SERVICE TIME: time to answer the call
how often a customer leaves



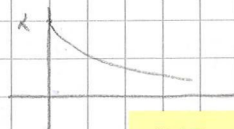
GOAL: what is the average number of customers in the system?

- what fraction of time do the customers spend in state 0, 1, 2...?
- How much time does it take to go from 0 to 5
i.e., how long it take to have someone in the waiting line?

=> EXP DIST

A random variable is exponentially distributed with parameter $\alpha > 0$ if its prob density function is

$$f(t) = \begin{cases} \alpha e^{-\alpha t} & t \geq 0 \\ 0 & \text{otherwise} \end{cases}$$



$$\begin{aligned} E[X] &= \int_0^{\infty} t f(t) dt = \int_0^{\infty} t \alpha e^{-\alpha t} dt \\ &= -t e^{-\alpha t} \Big|_0^{\infty} + \int_0^{\infty} e^{-\alpha t} dt \\ &= -t e^{-\alpha t} + \frac{1}{\alpha} e^{-\alpha t} \Big|_0^{\infty} = \frac{1}{\alpha} \end{aligned}$$

α = expected number of customers arriving in 1 hour
=> they arrive every $1/\alpha$ time
 $1/\alpha$ time between arrivals

EXP DIST:

prob dist of the time between events (in a POISSON PROCESS)

PROB DENSITY FUNCTION: function whose value at a given sample provides the likelihood that the value of the random variable would be equal to the sample

Events occur CONTINUOUSLY and INDEPENDENTLY at a constant average rate

$$P(X > a) = \int_a^\infty \lambda e^{-\lambda t} dt = -e^{-\lambda t} \Big|_a^\infty = e^{-\lambda a}$$

↑
 prob event occurs after a time

$$P(a < X < b) = \int_a^b \lambda e^{-\lambda t} dt = -e^{-\lambda t} \Big|_a^b = e^{-\lambda a} - e^{-\lambda b}$$

$$P(X > a+c | X > c) = P(X > a+b \text{ and } X > c) / P(X > c)$$

↑
 prob of something happening given that an event has already occurred

$$= e^{-\lambda(a+c)} / e^{-\lambda c} = e^{-\lambda a} = P(X > a)$$

⇒ the prob of a new arrival is always the same independently of how many there have been

⇒ lack of memory
 The events are independent on each other and on the time that takes out on the time that takes out on the next arrival

$$X \sim \exp(\lambda) \quad Y \sim \exp(\beta) \quad \text{independent}$$

$$Z = \min(X, Y)$$

X probit arrives
 Y probit arrives
 Z the first who arrives

$$P(Z > t) = P(X > t \text{ or } Y > t) = P(X > t)P(Y > t)$$

$$= e^{-\lambda t} e^{-\beta t} = e^{-(\lambda+\beta)t}$$

$$Z \sim \exp(\lambda+\beta)$$

$$X \sim \exp(\lambda) \quad Y \sim \exp(\beta)$$

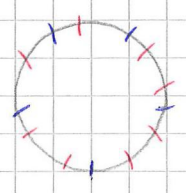
$$P(X < Y) = \int_0^\infty P(X < Y | X=t) f(t) dt$$

↑
 prob a probit arrives before first

$$= \int_0^\infty e^{-\lambda t} \lambda e^{-\lambda t} dt = \int_0^\infty \lambda e^{-(\lambda+\beta)t} dt$$

$$= -\frac{\lambda}{\lambda+\beta} e^{-(\lambda+\beta)t} \Big|_0^\infty = \frac{\lambda}{\lambda+\beta}$$

⇒ a probit arrives before another probit arriving of random time the prob of the first arrival being probit is $\lambda/(\lambda+\beta)$ (which from a over all possible arrivals $\lambda+\beta$)



POISSON DISTRIBUTION

OP with customers arriving with an $\exp(\lambda)$

i.e., λ customers arrive each hour

what is the prob of n events (arrivals) occurring in a certain interval of time?

$IP(n \text{ events in } [0, x])$?

$\Rightarrow$ POISSON DISTRIBUTION close relative of the $\exp$ dist

$$IP(0 \text{ events in } [0, x]) = IP(X \geq x) = e^{-\lambda x}$$

$$\begin{aligned} IP(1 \text{ event in } [0, x]) &= \int_0^x IP(0 \text{ events in } [0, x-t]) \lambda e^{-\lambda t} dt \\ &= \int_0^x e^{-\lambda(x-t)} \lambda e^{-\lambda t} dt \quad \begin{array}{l} \text{event happens at time } t \\ \text{memoryless} \end{array} \\ &= \int_0^x \lambda e^{-\lambda x} dt = \lambda t e^{-\lambda x} \Big|_0^x = \lambda x e^{-\lambda x} \end{aligned}$$

$$\begin{aligned} IP(2 \text{ events in } [0, x]) &= \int_0^x IP(1 \text{ event in } [0, x-t]) \lambda e^{-\lambda t} dt \\ &= \int_0^x \lambda(x-t) e^{-\lambda(x-t)} \lambda e^{-\lambda t} dt \\ &= \int_0^x \lambda^2 (x-t) e^{-\lambda x} dt \\ &= \lambda^2 \left(xt - \frac{1}{2} t^2 \right) e^{-\lambda x} \Big|_0^x = \lambda^2 \frac{x^2}{2} e^{-\lambda x} \end{aligned}$$

$$\begin{aligned} IP(3 \text{ events in } [0, x]) &= \int_0^x IP(2 \text{ events in } [0, x-t]) \lambda e^{-\lambda t} dt \\ &= \int_0^x \lambda^2 \frac{(x-t)^2}{2} e^{-\lambda(x-t)} \lambda e^{-\lambda t} dt \\ &= \int_0^x \lambda^3 \frac{(x-t)^2}{2} e^{-\lambda x} dt \\ &= \frac{\lambda^3}{6} (x-t)^3 e^{-\lambda x} \Big|_0^x = \lambda^3 \frac{x^3}{6} e^{-\lambda x} \end{aligned}$$

$$IP(n \text{ events in } [0, x]) = \frac{\lambda^n x^n}{n!} e^{-\lambda x}$$

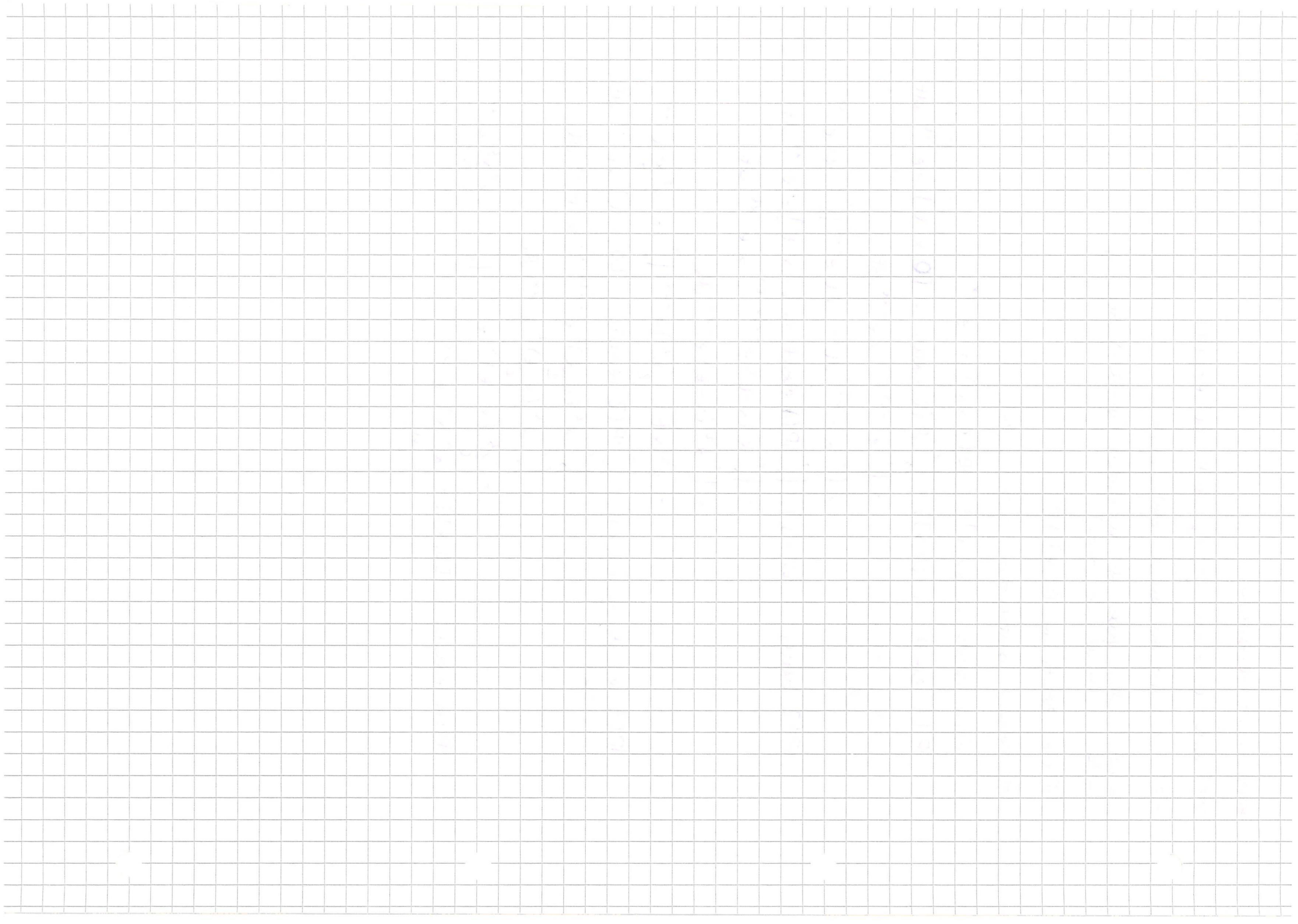
POISSON DIST

Prob of a given number of events occurring in a fixed interval of time if these events occur constantly and independently

$$IP(1 \text{ in } [0, x]) =$$

$$\int_0^x IP(0 \text{ in } [0, x] | \text{ arrival at } t) IP(X=t) dt$$

Event happens at t
and nothing has happened before



MEDICAL CLINIC

- Patients with FW arrive 10/hour (exp. dist)
- Physical injuries arrive 5/hour (exp. dist)
- FIND the PARAMETERS of the exp dist of FW and injury

$$\text{FW} : \text{exp}(\alpha) \quad \alpha = 10$$

$$\text{injury} : \text{exp}(\beta) \quad \beta = 5$$

- HOW LONG UNTIL THE FIRST FW?

$$\text{FW} : \text{exp}(10) \Rightarrow \text{IE}(\text{first FW}) = \frac{1}{10}$$

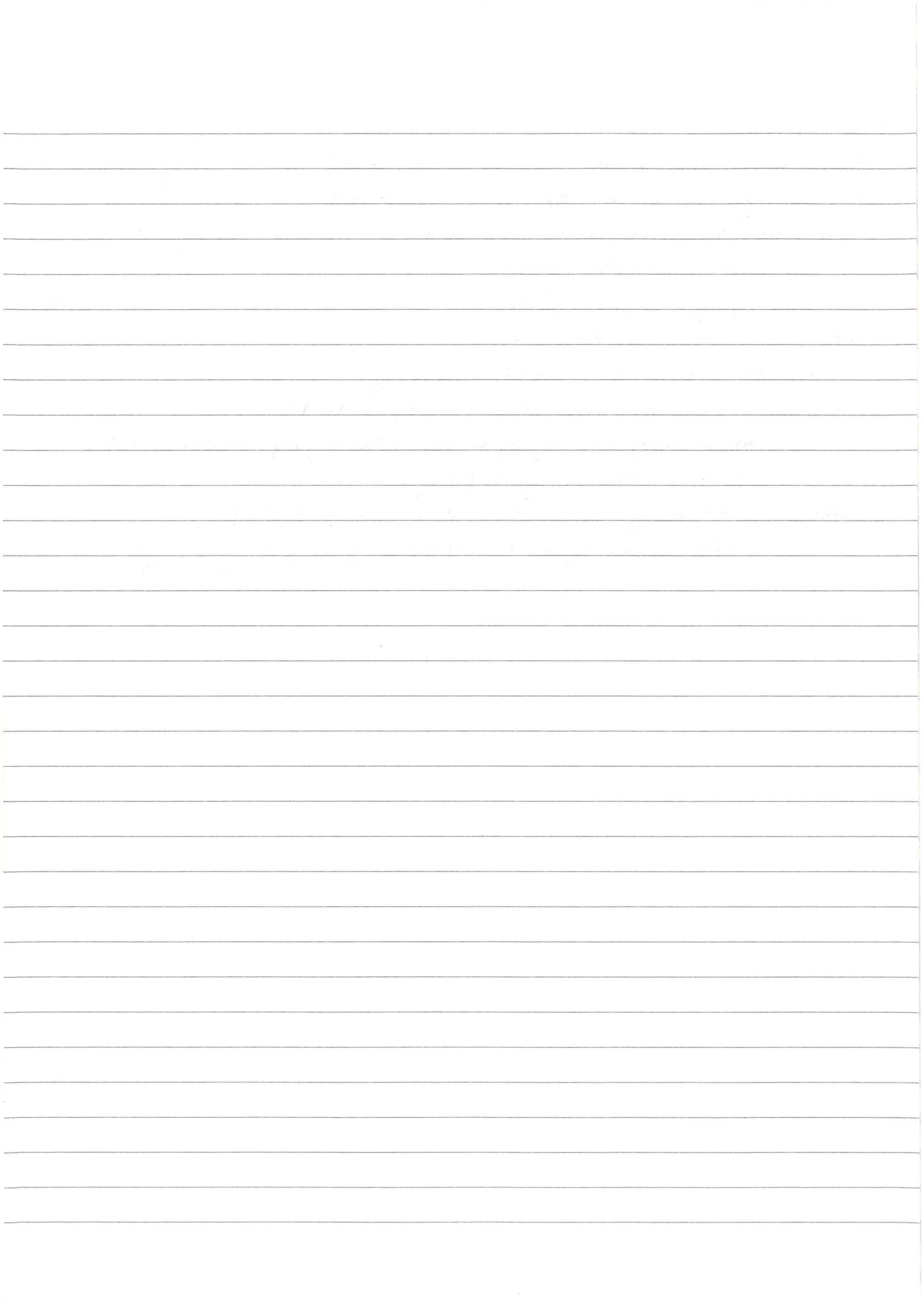
- HOW LONG UNTIL THE FIRST PATIENT (INJURY or FW)?

$$\text{ARRIVAL of FW or injury} = \text{min}\{\text{FW, injury}\} \sim \text{exp}(10+5)$$

$$\text{IE}(\text{first FW or injury}) = \frac{1}{15}$$

- WHAT is the PROBABILITY of the FIRST PATIENT HAVING the FW?

$$\text{IP}(\text{FW before injury}) = \text{IP}(\text{FW} < \text{injury}) = \frac{\alpha}{\alpha + \beta} = \frac{10}{15}$$



RECAP EXP and POISSON DISTRIBUTIONS

EXPONENTIAL DISTRIBUTION

$$\text{DENSITY FUNCTION: } f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

λ RATE PARAMETER

$E[x] =$ expected time between arrival

- λ arrivals every hour
 $\Rightarrow$ one arrival every $\frac{1}{\lambda}$ hours

EX: $\lambda = 30$

one arrival every $\frac{1}{30}$ hours = $\frac{60}{30}$ minutes = 2 minutes

$IP[x > a] =$ prob. of an event happening after a time

$IP[x > a+c \mid x > c] = IP[x > a] \rightarrow$ the prob of a new arrival is always the same independently of how many there have been before

$\downarrow$

prob of something happening after $a+c$ knowing that something happened already after c

LACK of MEMORY

$$Z = \min(X, Y) \quad X \sim \text{exp}(\lambda) \quad Y \sim \text{exp}(\beta)$$

$IP[Z > a] =$ prob of one arrival after a time, independently of the type
 $Z \sim \text{exp}(\lambda + \beta)$ - in total there are $\lambda + \beta$ arrivals

$IP[X < Y] =$ prob of arrival of type X before of arrival of type Y

$$= \frac{\lambda}{\lambda + \beta} = \frac{\text{FAVORABLE CASES}}{\text{POSSIBLE CASES}}$$

EXPONENTIAL DIST: prob dist of the time between events in a process in which events occur continuously and independently at a constant rate
(i.e. a POISSON PROCESS)

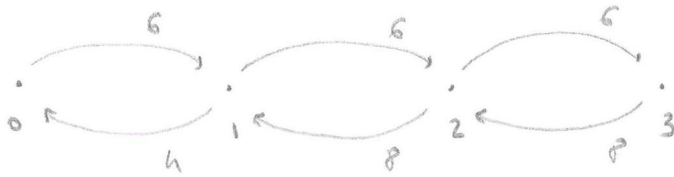
POISSON DISTRIBUTION

dist of the probs of a given number of events occurring in a fixed interval of time if these events occur with a constant rate and independently of the time since the last event

EXAMPLE

CALL CENTER (another one)

- 6 call per hour, $\exp(6)$
- 2 agents serving 4 customers per hour each, $\exp(4)$



How much time you spend in each state?

What is the fraction of time we spend in each state?

0

6 possible calls leaving from 0
with $\exp$ dist

$\Rightarrow$ we leave every $\frac{1}{6}$ hours
10 minutes

[$1E[X]$ $\exp$ dist]

[λ customers in
one hour

$\Rightarrow$ they arrive one
every $\frac{1}{\lambda}$ hours]

1

6 possible events leaving from 1 to 2
with $\exp$ dist

4 possible events leaving from 1 to 0
with $\exp$ dist

$\Rightarrow$ 10 events with $\exp$ dist
[$Z = \min(X, Y)$] we do not care if we change
state because a new customer
arrives or if an
agent leaves
we leave every $\frac{1}{10}$ hours
6 minutes

2

$6+8=14$ events leaving from 2
(to 3 and 1 respectively)

$\Rightarrow$ we leave every $\frac{1}{14}$ hours
 $\frac{30}{7}$ minutes

3

8 events leaving from 3

$\Rightarrow$ we leave every $\frac{1}{8}$ hours
 $\frac{15}{2}$ minutes

TRANSITION MATRIX

$$P_{01} = 1 \quad \text{only one link}$$

$$P_{10} = \frac{4}{4+6} = \frac{2}{5} \quad [IP[X < Y] = IP[0 \text{ before } 2]]$$

$$= \frac{\text{RATE to 0}}{\text{RATE to 0} + \text{RATE to 2}}$$

$$P_{12} = \frac{6}{4+6} = \frac{3}{5}$$

$$P_{21} = \frac{8}{8+6} = \frac{4}{7}$$

$$P_{23} = \frac{6}{8+6} = \frac{3}{7}$$

$$P_{32} = 1$$

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 2/5 & 0 & 3/5 & 0 \\ 0 & 4/7 & 0 & 3/7 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\pi = \pi P$$

$$\begin{cases} \pi_0 = \frac{2}{5} \pi_1 \\ \pi_1 = \pi_0 + \frac{4}{7} \pi_2 \\ \pi_2 = \frac{3}{5} \pi_1 + \pi_3 \\ \pi_3 = \frac{3}{7} \pi_2 \\ \pi_0 + \pi_1 + \pi_2 + \pi_3 = 1 \end{cases}$$

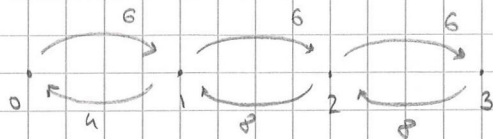
$$\pi = \frac{1}{56} (8, 20, 21, 9) \quad \text{supposing we spend equal time in each state}$$

=> we have to adjust to respect the actual times

ex: call center example but smaller

6 calls per hour with exp dist, that is, $\exp(6)$

2 agents serving 6 customers per hour each, $\exp(4)$



How much time you spend in each state?

0 10 min (6 calls per hour, one every 10 min)

1 6 min (6 calls + 4 served, one every 6 min)

2 $30/7$ min (6 calls + 8 served)

3 $15/2$ min (8 served)

$\Rightarrow$ we can treat the QS as a MC

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 2/5 & 0 & 3/5 & 0 \\ 0 & 4/7 & 0 & 3/7 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Follows from the properties of exp dist ($P(x < y)$)

$P(\text{customer leaving})$ $P(\text{new arrival})$

$$\pi = \pi P$$

$$\begin{cases} \pi_0 = \frac{2}{5} \pi_1 \\ \pi_1 = \pi_0 + \frac{4}{7} \pi_2 \\ \pi_2 = \frac{3}{8} \pi_1 + \pi_3 \\ \pi_3 = \frac{3}{7} \pi_2 \\ \pi_0 + \pi_1 + \pi_2 + \pi_3 = 1 \end{cases}$$

$$\pi_1 = \frac{2}{5} \pi_1 + \frac{4}{7} \pi_2 \quad \frac{3}{5} \pi_1 = \frac{4}{7} \pi_2 \quad \pi_2 = \frac{21}{20} \pi_1$$

$$\frac{21}{20} \pi_1 = \frac{3}{8} \pi_1 + \pi_3 \quad \pi_3 = \left(\frac{21}{20} - \frac{3}{8} \right) \pi_1 \quad \pi_3 = \frac{9}{20} \pi_1$$

$$\frac{2}{5} \pi_1 + \pi_1 + \frac{21}{20} \pi_1 + \frac{9}{20} \pi_1 = 1$$

$$(8 + 20 + 21 + 9) \pi_1 = 20$$

$$\pi_1 = \frac{20}{58}$$

$$\pi = \left(\frac{2}{58}, \frac{20}{58}, \frac{21}{58}, \frac{9}{58} \right)$$

(supposing we spend equal time in each state)

$\Rightarrow$ we can adjust this to reflect the actual times as you did already for MC with different times

10	6	30	15
$\frac{140}{16}$	$\frac{84}{14}$	$\frac{60}{14}$	$\frac{105}{16}$

$$\frac{140 \cdot 8}{1120} + \frac{84 \cdot 20}{1680} + \frac{21 \cdot 60}{1260} + \frac{9 \cdot 105}{945} = 5205$$

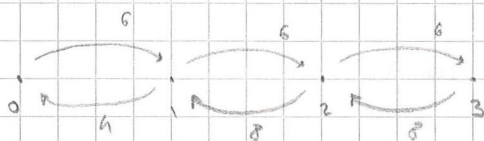
$$\pi = \left(\frac{1120}{5205}, \frac{1680}{5205}, \frac{1260}{5205}, \frac{945}{5205} \right) \text{ reflects the working hours}$$

(actual stationary distribution of the QS)

Is there a faster way to do this?

"RATE IN" = "RATE OUT" PRINCIPLE

=> automatically gives the stationary dist of the QS



IN = OUT

$$0 \quad 4\pi_1 = 6\pi_0$$

$$1 \quad 6\pi_0 + 8\pi_2 = 6\pi_1 + 4\pi_1$$

$$2 \quad 6\pi_1 + 8\pi_3 = 6\pi_2 + 8\pi_2$$

$$3 \quad 6\pi_2 = 8\pi_3$$

$$\pi_0 + \pi_1 + \pi_2 + \pi_3 = 1$$

$$\pi_0 = \frac{2}{3}\pi_1$$

$$\pi_2 = \frac{3}{4}\pi_1$$

$$\pi_3 = \frac{3}{4}\pi_2 = \frac{9}{16}\pi_1$$

$$\frac{2}{3}\pi_1 + \pi_1 + \frac{3}{4}\pi_1 + \frac{9}{16}\pi_1 = 1$$

$$(32 + 48 + 36 + 27)\pi_1 = 48$$

$$\pi_1 = \frac{48}{143}$$

$$\pi = \left(\frac{32}{143}, \frac{48}{143}, \frac{36}{143}, \frac{27}{143} \right) = (0.22, 0.33, 0.25, 0.18)$$

Does this look the same as the one we obtained with the MC?

$$\pi = \left(\frac{1120}{5205}, \frac{1680}{5205}, \frac{1260}{5205}, \frac{945}{5205} \right) = (0.22, 0.33, 0.25, 0.18)$$

(with small differences due to rounding)

Fraction of time spent in state 0, 1, 2, 3

• what is the average number of columns in the item?

$$0 \cdot 0.22 + 1 \cdot 0.33 + 2 \cdot 0.25 + 3 \cdot 0.18 = 1.37$$

$\begin{matrix} 0 & 0.33 & 0.5 & 0.54 \end{matrix}$

How much time do we need to go from state 0 to state 3 (in minutes)?

How long it takes for the piece to be fixed?

E_x = time to go from x to 3

$$\begin{cases} E_0 = 10 + E_1 \\ E_1 = 6 + \frac{2}{3}E_0 + \frac{3}{5}E_2 \\ E_2 = \frac{30}{7} + \frac{4}{7}E_1 + \frac{3}{7}E_1 \\ E_3 = 0 \end{cases}$$



$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 2/3 & 0 & 3/5 & 0 \\ 0 & 4/7 & 0 & 1/7 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$E_1 = E_0 - 10$$

$$E_0 - 10 = 6 + \frac{2}{3}E_0 + \frac{3}{5}E_2$$

$$E_0 - \frac{80}{3} = \frac{30}{7} + \frac{4}{7}(E_0 - 10)$$

$$E_2 = \frac{5}{3} \left(\frac{3}{5}E_0 - 16 \right) = E_0 - \frac{80}{3}$$

$$\frac{3}{7}E_0 = \frac{80}{3} + \frac{30}{7} + \frac{40}{7}$$

$$9E_0 = 560 + 90 + 120$$

$$E_0 = \frac{530}{9} \approx 58.9$$

